SR-TE Yang modules

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Overview for IETF-127

Adopted Drafts

- Spring Yang documents
  - RFC9020
  - draft-ietf-spring-srv6-yang-base-02.txt
  - draft-ietf-spring-srv6-yang-static-00,
  - ID: draft-gong-spring-sr-policy-group-yang-04
- SR-TE IDR Yang drafts
  - draft-lin-idr-bgp-sr-policy-yang
- SR-TE BGP-LS Yang drafts
  - draft-Aravind-idr-bgp-ls-sr-policy-yang

draft-ietf-spring-srv6-yang-base-02

SR content

- Configuration
- Operational state
- Notifications

SR building blocks

- SID: LOC:FUNCT:ARG
  - LOC – SID-Block:Nod-ID
  - LBL: LNL:FL:AL
- SID
  - Uncompressed SID
  - Compressed SID: (CSID)
    - Next-CSID – LBL: 32, LNL:16, FL:16
    - Replace CSID - LBL: 32, LNL:32, FL:32
    - SID space: Global ID Block (GIB) + Local ID Block (LIB)
    - LIB space: Dynamic LIB, Explicit LIB, Reserved LIB

SR structures - config

RFC9020

Module: ietf-segment-routing-mpls

augment /rt:routing/sr:segment-routing:

+--rw srmpls [RFC9020]

 +--rw bindings

 | …

 +--rw srgb

 | …

 +--rw srlb

 | …

[draft-ietf-spring-srv6-yang-base]

Module: ietf-srv6-base

augment /rt:routing/sr-segment-routing

+--rw srv6

 +--rw enable

 | …

 +--rw encapsulation

 | …

 +--rw sid-formats

 | …

 +--rw locators

 | …

 +--rw-sids

SR structures – operational state

RFC 9020

Module: ietf-segment-routing-mpls

augment /rt:routing/sr:segment-routing:

+--rw srmpls

| ….

+--ro label-blocks* []

| ….

+--ro sid-db

draft-ietf-spring-srv6-yang-base

Module: ietf-srv6-base

augment /rt:routing/sr-segment-routing

+--rw srv6

| …

+--ro node-capabilities

| …

+--rw sid-formats (sids by format)

|

+--rw locators (by name, remote prefix)

|

+--rw sids

SR structures – notification

RFC 9020

Module: ietf-segment-routing-mpls

Notifications:

```
+---n segment-routing-srgb-collision
 |    +ro srgb-collisons
 |     ….
 +---n segment routing-global-sid-collison
 | ….
 +---n segment-routing-index-out-of-range
```

draft-ietf-spring-srv6-yang-base

Module: ietf-srv6-base

notification

```
+---n srv6-locator-status-event
 |    +--ro operational-status?
 |    +-- ro locator?
+---n srv6-sid-collision-event
 |     +--ro-sid?
 |     +--ro existing
 |      ….
 |     +ro requested
```

SR-TE Config + operational

draft-ietf-spring-sr-policy-yang
Module: ietf-sr-policy

```
augment /rt:routing:
+--rw segment-routing
   +--rw traffic-engineering
      +--rw computation-templates
      |  ...
      +--rw optimization-templates
      | ...
      +--rw attributes
      |  ...
      +--rw policies
         +--rw –(ro name, headend, color, EP
                  admin-state, priorit, BW requested)
         +--ro - BSID, SRv6-SIDs, counters,
                 CP (explicit, dynamic, composite) DB, best
```

[draft-lin-idr-bgp-sr-policy-yang-00]
Module: ietf-bgp-sr-policy

```
augment /rt:routing/rt:control-plane-protocols
      /rt:control-plane-protocol/bgp:bgp/bgp:global
      /bgp:afi-safis/bgp:afi-safi
      +--rw sr-policy
         +--rw prefix-limit
         | …
         +--rw propagate-to-e-bgp?
         +--rw ignore-unsupported-subtlvs?
      +--ro sr-policy
         +--ro loc-rib
          | ....
          +ro neighbors
          ....
```

SR-TE Notifications

draft-ietf-spring-sr-policy-yang

Module: ietf-sr-policy

augment /rt:routing:

+----n sr-policy-oper-state-change-event

| .....

+----n sr-policy-candidate-path-change-event

|....

+----n sr-policy-binding-sid-unavailable-event

| ....

+----n sr-policy-candidate-path-binding-sid-mismatch

|...

[draft-lin-idr-bgp-sr-policy-yang-00]

Module: ietf-srv6-base

? None

SR-TE groups

Draft-gong-spring-sr-policy-group-yang

Module: ietf-sr-policy

```
Segment-routing
  Traffic-engineering
       + rw attributes
       |  + ......
       +--rw  policies
       |   +--rw--policy* [color endpoint]
       |      ......
       +--rw sr-policy-groups
           +--rw groups* [group-color]
```

[**draft-gong-spring-sr-policy-group-yang-04]**

Module defined: ietf-sr-policy-group

```
Augment /rt:routing
   +--rw sr-policy-groups
       +--rw groups* [group-color]
           +--rw group-color
           +--rw name?
           +--rw description
           +--rw service-distringuish-method
           +--rw forward-based-dscp
           | …  (config)
           +--rw dscp color-mapping-entries
           |…(config)
           +--rw forward-based-dot1q
           | … (config)
          +--rw forward-based-service-class
```

SR-TE BGP-LS

[**draft-aravind-idr-bgp-ls-sr-policy-yang]**

Module defined: ietf-bgp-ls-sr-policy

Augment /rt:routing/rt:control-plane-protocols
/rt:control-plane-protocol/bgp:bgp/bgp:global
/bgp:afi-safis/bgp:afi-safi/bgp-ls:link-state
/bgp-ls:bgp-ls-topology/bgp-ls:instances
/bgp-ls:instance:
+--ro sr-policy-candidate-paths
+---ro candidate-path* [as bgp-router-id
member-as ipv4-router-id
ipv6-router-id protocol-origin endpoint
color originator-as originator-address
descriminator]
| …. (Operational data)

[**draft-Aravind-idr-bgp-ls-policy-yang**

Module defined:

iana-bgp-ls-sr-policy-types

Typedefs

- protocol-origin
- segment-type
- Metric-type

SRv6 Next Hop Additions

[**draft-lin-rtgwg-srv6-nexthop-yang-03.txt]**

Module defined: ietf-srv6-nexthop

Augment /rt:routing/rt:control-plane-protocols
/rt:control-plane-protocol/rt:staticroutes/v4ur:ipv4
/v4ur:route/v4ur:next-hop/v4ur:next-hop-options

```
+--:(v4ur: srv6-next-hop-list)
   +---rw v4ur:srv6-next-hop* [index]
      +--rw v4ur:index  string
      +--rw v4ur-:srv6-te
         +--rw v4ur:color sr-policy-types:color-type
         |… (color, endpoint, srv6-sid)
         +--rw v4ur: srv6-be
            +rw 4vur:srv6-sid inet:ipv6-prefix
```

Augment /rt:routing/rt:control-plane-protocols
/rt:control-plane-protocol/rt:static-routes/ v6ur:ipv6
/v6ur:route/v6ur:next-hop/v6ur:next-hop-options:

```
+--:(v4ur: srv6-next-hop-list)
   +---rw v4ur:srv6-next-hop* [index]
      +--rw v4ur:index  string
      +--rw v4ur-:srv6-te
         +--rw v4ur:color  sr-policy-types:color-type
         |… (color, endpoint, srv6-sid)
         +--rw v4ur: srv6-be
            +rw 4vur:srv6-sid inet:ipv6-prefix
```

SRv6 Next Hop Additions (operational)

[**draft-lin-rtgwg-srv6-nexthop-yang-03.txt]**

Module defined: ietf-srv6-nexthop

```
augment /rt:routing/rt:ribs/rt:rib/rt:routes/rt:route
        /rt:next-hop/rt:next-hop-options:
+--:(srv6-next-hop-list)
     +---ro srv6-next-hop-list
          +--ro srv6-next-hop*
           |     +--ro srv6-te
                 | …. (color, endpoint, srv6-sid)
           |     +--ro-srv6-be
                       +--ro srv6-sid
```

```
augment /rt:routing/rt:ribs/rt:rib/rt:active-route/rt:output
        /rt:route/rt:next-hop/rt:next-hop-options:
+--:(srv6-next-hop-list)
    +---ro srv6-next-hop-list)
         +--ro srv6-next-hop*
              +--ro srv6-te
               | .. (color, endpoint, SRv6-sid)
              +---ro srv6-be
                   +--ro srv6-sid
```

SR TE conclusions

- IDR drafts
  - draft-lin-idr-bgp-sr-policy-yang
  - draft-aravind-idr-bgp-ls-sr-policy-yang/
- Status:
  - No conflict with Spring
  - Alignment with basic bgp-model
  - Question: Do we need more BGP-LS augments to base bgp-module?
- Next Steps
  - Present at IETF-127 (with chairs introduction)
  - Interim before IETF-128
  - Where is NRP-ID in next Hop tables in Yang?